



Series and Sigma Notation

The sum of the first four terms of the sequence $1, 3, 5, 7, \dots$ is represented by the sum $1 + 3 + 5 + 7$.

The expression $1 + 3 + 5 + 7$ is an example of a series.

Definition

A **series** is the sum of consecutive terms of a sequence.

The notation S_n represents the sum of the first n terms of a series.

Sequence: $2, 5, 8, 11, 14, \dots$

Corresponding Series: $2 + 5 + 8 + 11 + 14 + \dots$

$$S_1 = 2$$

$$S_2 = 2 + 5 = 7$$

$$S_3 = 2 + 5 + 8 = 15$$

$$S_4 = 2 + 5 + 8 + 11 = 26$$

$$S_5 = 2 + 5 + 8 + 11 + 14 = 40$$

$\dots$ and so on. When a series has many terms, it can be convenient to use sigma notation, also known as summation notation. Sigma notation uses the Greek capital letter sigma: Σ .

► **Think About It** The sum of the first n terms of a series is called the n th partial sum.

Finding a Sum in Sigma Notation

Definitions

The sum S_n of the first n terms of a sequence can be represented as

$$S_n = \sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n.$$

S_n is equivalent to $\sum_{i=1}^n a_i$, which is read as “the sum from 1 to n of a_i .”

In this notation, i is called the **index**, n is called the **upper limit**, and 1 is the **lower limit**.

► **Think About It** The lower limit of a series in sigma notation is not always equal to 1. The lower limit can be any whole number less than or equal to n .

To find the sum of a series given in sigma notation, begin by writing it in expanded form. You can then find the sum by adding the terms of the series.

EXAMPLE 1

Write the series in expanded form and find the sum.

A $\sum_{i=1}^4 6i$

SOLUTION

$$\begin{aligned}\sum_{i=1}^4 6i &= 6 \cdot 1 + 6 \cdot 2 + 6 \cdot 3 + 6 \cdot 4 && \text{Expanded form} \\ &= 6 + 12 + 18 + 24 && \text{Simplify.} \\ &= 60\end{aligned}$$

B $\sum_{k=1}^6 k^2$

SOLUTION

$$\begin{aligned}\sum_{k=1}^6 k^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 && \text{Expanded form} \\ &= 1 + 4 + 9 + 16 + 25 + 36 && \text{Simplify.} \\ &= 91 \quad \blacksquare\end{aligned}$$

Using Sigma Notation to Describe a Series

EXAMPLE 2

Write the series using sigma notation.

A $4 + 9 + 14 + 19 + 24 + 29 + 34$

SOLUTION

This series is arithmetic, with a first term of 4 and a common difference of 5. The iterative rule for the related sequence is $a_i = 5i - 1$. There are $n = 7$ terms in the series.

The answer is $\sum_{i=1}^7 (5i - 1)$. You can check the answer by writing it in expanded form.

► **Remember** The iterative rule for an arithmetic sequence is $a_n = a_1 + (n - 1)d$, and the iterative rule for a geometric sequence is $a_n = a_1 \cdot r^{n-1}$.

B $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

SOLUTION

This series is geometric, with a first term of 1 and a common ratio of $\frac{1}{2}$.

The iterative rule for the related sequence is $a_i = \left(\frac{1}{2}\right)^{i-1}$. There are $n = 4$ terms in the series.

The answer is $\sum_{i=1}^4 \left(\frac{1}{2}\right)^{i-1}$. You can check the answer by writing it in expanded form. ■